

Executive Overview: Angular Manifold Routing

Artificial intelligence systems have grown dramatically in capability, but this progress has come with a cost. Modern large-scale AI models require enormous computational infrastructure. Training and running these systems can demand thousands of GPUs, vast energy consumption, and complex distributed systems.

A large portion of this cost arises because neural networks typically perform large volumes of computation even when much of it contributes little to the final result. In most architectures, every input activates nearly the entire model. This means computational resources are spent uniformly, even though the useful signal within the data is often structured and clustered.

Researchers have therefore been exploring methods that allow models to perform computation selectively. If a model could determine which parts of itself are relevant for a particular input, it could activate only those components and skip the rest. This concept is known broadly as sparse or conditional computation.

The Central Idea

Angular Manifold Routing introduces a geometric mechanism that naturally produces sparse computation. Instead of relying on complex learned gating networks, the system organizes token embeddings according to their direction in a normalized vector space.

When embeddings are normalized, they lie on the surface of a unit hypersphere. The routing system divides this hypersphere into angular sectors. Each token is assigned to the sector closest to its direction. Tokens with similar semantic meaning tend to point in similar directions, which causes them to cluster into shared routing buckets.

Observed Routing Behavior

Experiments show that tokens do not distribute evenly across available routing sectors. Instead, they cluster strongly. The number of routing buckets that become active follows a consistent scaling pattern:

$$E(K) \approx c \cdot K^{0.572}$$

Where: K represents the number of routing sectors available, $E(K)$ represents the number of sectors actually used.

Because the exponent is less than 1, the number of active routing buckets grows more slowly than the number of available buckets. This produces a sublinear scaling effect. Even when thousands of routing sectors exist, only a fraction of them are active at any given time.

Available Buckets (K)	Typical Active Buckets (Observed)
100	~40
1,000	~150
5,000	~350

Practical Implications

1. **Reduced Computation** Because only active routing buckets perform computation, the model performs significantly less work overall.
2. **Improved Hardware Efficiency** Clustering tokens into fewer buckets improves memory locality and reduces random memory access patterns.
3. **Better Cache Performance** Experiments show approximately 2.5× fewer cache misses in simulated hardware environments.
4. **Lower Effective Routing Cost** Across multiple experiments, the approach demonstrated roughly 3–5× reductions in effective routing cost.

Why This Happens

The phenomenon appears to arise from geometry rather than learned gating behavior. Because embeddings representing related information tend to occupy nearby directions in vector space, angular partitioning naturally groups similar tokens together.

This means sparsity is not imposed artificially by the model. Instead, it emerges as a property of the embedding geometry itself.

What Has Been Verified

Experiments have verified several important properties of the routing mechanism:

- The scaling behavior appears consistently across multiple routing capacities.
- The effect persists across different vector normalization schemes.
- Hardware proxy tests show meaningful reductions in memory traffic and routing cost.
- The phenomenon appears driven purely by angular geometry rather than radial partitioning.

What Comes Next

The current experiments were conducted using controlled routing environments and diagnostic models. The next phase of research will integrate Angular Manifold Routing into full neural network architectures, including transformer models used in large language systems.

These experiments will determine whether the same scaling behavior appears in production-scale AI models. If the effect persists, it could significantly reduce the computational cost of large AI systems.

In summary, Angular Manifold Routing introduces a geometry-based routing mechanism that naturally produces sparse computation. By organizing embeddings according to their direction on a normalized hypersphere, the system allows many tokens to share compute pathways.

The resulting sublinear routing law suggests a new approach to designing efficient large-scale AI systems, potentially enabling substantial improvements in hardware efficiency and scalability.